

ELECTIVE-7.3

URA733: INDUSTRY 4.0 COMPLIANT PRODUCT DESIGN AND MANUFACTURING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course is designed to introduce the students to the integrated and interdisciplinary product design philosophy considering social, historical, functional, technological and physiological aspects of product and industrial design to enhance their abilities to use the tools and techniques to translate their concepts and imagination into useful products through skill-building in graphic representation and prototyping.				
Syllabus Introduction to Product Design: Product definition, types of product development processes; Concept of Product Life Cycle; identifying customer needs and validation; concept generation; concept selection; product architecture and industrial design; human factors & system information input; Creative thinking. <i>Case studies (on workplace design, human factors application, human errors – accidents and safety)</i> Integrative Design Studio: Concept evaluation: black box approach, system analysis viewpoints, absolute and relative, decision process approach. The creative process: what, how and why to design and make a product? How to proceed from an idea? Possible ways to approach a problem, Working with cross- disciplinary projects. <i>Students need to actively: question, observe, visualize, play, fail, reframe, experiment, fabricate, and reflect. Short group assignments to introduce a studio based methods for idea generation, integrating research, and documenting discoveries in mechanical product design.</i> Portfolio, Publication and Dynamic Media: Methods for documenting and communicating design ideas and achievements to different audiences through presentations or creating promotional materials in digital and print format. Space and materiality: Types of spaces influencing product design; Concept of personal space and shared space; Review of materials and processes commonly used by product designers and manufacturers at various scales of production; Learning through failure: trial and error approach to finding out about material properties, role of malleability, weight, texture, durability, and ways to engage all the senses to create three- dimensional prototypes; Social and ecological impacts of the materials used for product design. Product Development Process & Methodologies: Translating an idea to a three-dimensional world using physical construction; Effect of force, resistance and gravity; Bottom-up and top-down design approach, front loading design workflow, design in context, modular design, product reliability, mortality curve, design for assembly and manufacturability; roles of prototype, need for rapid-prototyping and outsourced				

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fabrication; Techniques of product evaluation, design for product life cycle and need of Product Life Cycle Management System.

Laboratory Work (if applicable)

1. 3D CAD modelling of the product design to integrate the IoT enabled digital sensors into the CAD models. The real world sensor data can be taken from the test bed and can be refined and analysed to make better real time design decisions. The software like **Cero Product Insight extension or Autodesk Research** and **Autodesk viewer applications** can be used.

2. **Live project:** The participation in the live project will provide the students an opportunity to sharpen their prior learning skills by employing design processes in iterative, active, and analytical ways utilizing prototyping, testing, and comparative research. In such a design project the students are required to:

- a. Design a solution for an interdisciplinary live project that uses the **integration of the digital sensors for product design, evaluation and deployment** for remote access and data collection.
- b. The student team may include participants from other disciplines in the multidisciplinary team projects with a high degree of project closure and advanced presentation.
- c. Select a project from various domains but not limited to automobile, aero-space, pressure vessel, pump, compressor, pipeline, domestic appliances, Indian kitchen, hardware accessories used in building construction and modular furniture etc.
- d. Formulate a concise ***Problem definition*** and ***objective formulation, history and technological progress in the field*** and ***state of art, design and fabrication of the final product with detailed production drawing, engineering tolerances*** in light of the **regulatory compliances** and **engineering standards**.
- e. Showcase the final project work outcome and the completed proof of concept(s) models to the experts and peers from within the department or outside the department of study. The projects/s undertaken during the course must be documented as a **print portfolio, collection of short videos** as well as digital formats including **basic web design, social media communication**, etc.

Course Learning Objectives (CLO)

The students will be able to:

1. compile the design requirements as per the customer need and analysis, create and evaluate the detailed design requirements as well as the production plan to meet the estimated cost targets.
2. design a given product considering the social, human and technological aspects to meet the underlined functional and operating requirements.

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